

PADHAISHURU

GATE Preparation Platform · padhaishuru.com

TREND-BASED MOCK PAPER

GATE Biotechnology

Mock Paper — 04

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Biotechnology
Paper:	Mock 04
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

Unofficial Practice Material — Not affiliated with or endorsed by GATE/IIT.

Prepared by PadhaiShuru (padhaishuru.com) for practice purposes only.

Section A: General Aptitude

Q1. [1-NAT] [1 mark]

Choose similar to 'catalyst':

(Numerical Answer Type — Enter the numerical value)

Section B: Biotechnology

Q2. [1-NAT] [1 mark]

Normal distribution mean = median = ?

(Numerical Answer Type — Enter the numerical value)

Q3. [1-NAT] [1 mark]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

(Numerical Answer Type — Enter the numerical value)

Q4. [1-MCQ] [1 mark]

Number of spanning trees in complete graph K_4 :

- (A) 4
 - (B) 8
 - (C) 16
 - (D) 12
-

Q5. [1-MCQ] [1 mark]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q6. [1-MCQ] [1 mark]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q7. [1-MCQ] [1 mark]

Which of the following is a probability generating function for $\text{Poisson}(\lambda)$?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{-\lambda}(1+z)$

Q8. [1-MCQ] [1 mark]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$
- (D) $(1, 2, 3)$

Q9. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q10. [2-NAT] [2 marks]

The rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$ is:

(Numerical Answer Type — Enter the numerical value)

Q11. [2-MCQ] [2 marks]

$d/dx[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q12. [2-NAT] [2 marks]

The Laplace transform of $t e^{-2t}$ is:

(Numerical Answer Type — Enter the numerical value)

Q13. [2-NAT] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

(Numerical Answer Type — Enter the numerical value)

Q14. [2-MCQ] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68

- (B) 0.32
- (C) 0.95
- (D) 0.16

Q15. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q16. [2-MCQ] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q17. [2-NAT] [2 marks]

Laplace transform of $e^{(-3t)}u(t)$ is:

(Numerical Answer Type — Enter the numerical value)

Q18. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q19. [2-MCQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C * x$
- (B) $y = C/x$
- (C) $y = C * x^2$
- (D) $x^2 + y^2 = C$

Q20. [2-NAT] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

(Numerical Answer Type — Enter the numerical value)

Q21. [2-NAT] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} * \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ equals:

(Numerical Answer Type — Enter the numerical value)

Q22. [2-MCQ] [2 marks]

The area of a circle of radius r is:

- (A) $\pi * r^2$
- (B) $2 * \pi * r$
- (C) $\pi * r$
- (D) $2 * \pi * r^2$

Q23. [2-MCQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q24. [2-MCQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q25. [2-MCQ] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q26. [2-MCQ] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q27. [2-MCQ] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q28. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q29. [2-MCQ] [2 marks]

Consider $f(x) = |x|$ on $[-1, 1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q30. [2-MCQ] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q31. [1-MCQ] [1 mark]

Mendelian dihybrid cross (both heterozygous) ratio:

- (A) 3:1
- (B) 9:3:3:1
- (C) 1:2:1
- (D) 1:1:1:1

Q32. [1-NAT] [1 mark]

Competitive enzyme inhibition: K_m

(Numerical Answer Type — Enter the numerical value)

Q33. [1-MCQ] [1 mark]

DNA replication in eukaryotes occurs in:

- (A) Cytoplasm
- (B) Nucleus
- (C) Mitochondria only
- (D) G1 phase

Q34. [1-NAT] [1 mark]

Central dogma: reverse transcription is:

(Numerical Answer Type — Enter the numerical value)

Q35. [1-NAT] [1 mark]

Photosynthesis light reactions produce:

(Numerical Answer Type — Enter the numerical value)

Q36. [2-MSQ] [2 marks]

Mendelian dihybrid cross (both heterozygous) ratio:

- (A) 3:1
- (B) 9:3:3:1
- (C) 1:2:1
- (D) 1:1:1:1

Q37. [2-MCQ] [2 marks]

Central dogma: reverse transcription is:

- (A) RNA->DNA
- (B) DNA->RNA
- (C) Protein->DNA
- (D) RNA->Protein

Q38. [2-MCQ] [2 marks]

DNA replication in eukaryotes occurs in:

- (A) Cytoplasm

- (B) Nucleus
- (C) Mitochondria only
- (D) G1 phase

Q39. [2-NAT] [2 marks]

Competitive enzyme inhibition: K_m

(Numerical Answer Type — Enter the numerical value)

Q40. [2-MCQ] [2 marks]

DNA replication in eukaryotes occurs in:

- (A) Cytoplasm
- (B) Nucleus
- (C) Mitochondria only
- (D) G1 phase

Q41. [2-NAT] [2 marks]

Mendelian dihybrid cross (both heterozygous) ratio:

(Numerical Answer Type — Enter the numerical value)

Q42. [2-MSQ] [2 marks]

Central dogma: reverse transcription is:

- (A) RNA->DNA
- (B) DNA->RNA
- (C) Protein->DNA
- (D) RNA->Protein

Q43. [1-MCQ] [1 mark]

Galvanic cell converts:

- (A) Electrical to chemical
- (B) Chemical to electrical
- (C) Heat to work
- (D) Mechanical to electrical

Q44. [1-MCQ] [1 mark]

pH of 10^{-8} M HCl approx:

- (A) 6
- (B) 7
- (C) 8
- (D) 6.98

Q45. [1-MCQ] [1 mark]

Strongest acid:

- (A) HCl
- (B) HNO₃
- (C) H₂SO₄
- (D) HClO₄

Q46. [1-MCQ] [1 mark]

First-order half-life with $k=0.693 \text{ min}^{-1}$:

- (A) 1 min
- (B) 0.693 min

- (C) 1.44 min
- (D) 0.5 min

Q47. [2-MCQ] [2 marks]

$K_p = K_c(RT)^{\Delta n}$. For $N_2 + 3H_2 \rightleftharpoons 2NH_3$, $\Delta n =$

- (A) 2
- (B) -2
- (C) 1
- (D) -1

Q48. [2-NAT] [2 marks]

pH of 10^{-8} M HCl approx:

(Numerical Answer Type — Enter the numerical value)

Q49. [2-MCQ] [2 marks]

$K_p = K_c(RT)^{\Delta n}$. For $N_2 + 3H_2 \rightleftharpoons 2NH_3$, $\Delta n =$

- (A) 2
- (B) -2
- (C) 1
- (D) -1

Q50. [2-MCQ] [2 marks]

pH of 10^{-8} M HCl approx:

- (A) 6
- (B) 7
- (C) 8
- (D) 6.98

Q51. [1-NAT] [1 mark]

DNA stands for:

(Numerical Answer Type — Enter the numerical value)

Q52. [1-NAT] [1 mark]

Law of segregation describes:

(Numerical Answer Type — Enter the numerical value)

Q53. [1-MCQ] [1 mark]

mRNA processing:

- (A) Transcription only
- (B) Capping, splicing, poly-A
- (C) Translation only
- (D) Replication

Q54. [1-NAT] [1 mark]

Linked genes show:

(Numerical Answer Type — Enter the numerical value)

Q55. [1-MCQ] [1 mark]

ABO blood group has how many alleles?

- (A) 2

- (B) 3
- (C) 4
- (D) 6

Q56. [1-NAT] [1 mark]

DNA B-form base pairs per turn:

(Numerical Answer Type — Enter the numerical value)

Q57. [1-MCQ] [1 mark]

AaBb x aabb. Genotypic ratio:

- (A) 1:1:1:1
- (B) 3:1
- (C) 9:3:3:1
- (D) 1:2:1

Q58. [2-MSQ] [2 marks]

DNA B-form base pairs per turn:

- (A) 10
- (B) 10.4
- (C) 12
- (D) 8

Q59. [2-MCQ] [2 marks]

mRNA processing:

- (A) Transcription only
- (B) Capping, splicing, poly-A
- (C) Translation only
- (D) Replication

Q60. [1-MCQ] [1 mark]

Amino acids linked by:

- (A) Peptide bond
- (B) Hydrogen bond
- (C) Ionic bond
- (D) None

Q61. [1-NAT] [1 mark]

Binary fission in:

(Numerical Answer Type — Enter the numerical value)

Q62. [1-MCQ] [1 mark]

Batch process is:

- (A) Continuous
- (B) Intermittent
- (C) Steady
- (D) None

Q63. [1-MCQ] [1 mark]

Antibody produced by:

- (A) T cells
- (B) B cells

- (C) Macrophages
 - (D) None
-

Q64. [1-MCQ] [1 mark]

BLAST is for:

- (A) Sequence alignment
 - (B) Structure prediction
 - (C) Phylogeny
 - (D) None
-

Q65. [1-MCQ] [1 mark]

Chromosome number in daughter:

- (A) Halved
 - (B) Same
 - (C) Doubled
 - (D) Variable
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1NAT	B	1
2	1NAT	C	1
3	1NAT	A	1
4	1NAT	A	1
5	1MCQ	A	1
6	1NAT	A	1
7	1MCQ	B	1
8	1MCQ	B	1
9	1MCQ	A	1
10	1MCQ	B	1
11	1MCQ	B	1
12	1MCQ	B	1
13	1NAT	B	1

14	1NAT	C	1
15	1MCQ	C	1
16	1MCQ	B	1
17	1NAT	A	1
18	1MCQ	C	1
19	1MCQ	B	1
20	1MCQ	D	1
21	1NAT	A	1
22	1MCQ	A	1
23	1MCQ	D	1
24	1MCQ	A	1
25	1MCQ	B	1
26	1MCQ	A	1
27	1NAT	B	1
28	1NAT	B	1

29	1MCQ	B	1
30	1MCQ	A	1
31	2MCQ	B	2
32	2MCQ	C	2
33	2NAT	A	2
34	2MCQ	C	2
35	2NAT	A	2
36	2NAT	A	2
37	2MCQ	A	2
38	2MCQ	B	2
39	2MCQ	A	2
40	2NAT	A	2
41	2MCQ	A	2
42	2MCQ	B	2
43	2NAT	C	2

44	2NAT	A	2
45	2MCQ	A	2
46	2MCQ	A	2
47	2MCQ	A	2
48	2MCQ	A	2
49	2MCQ	B	2
50	2MCQ	B	2
51	2MSQ	B	2
52	2MCQ	A	2
53	2NAT	D	2
54	2MCQ	B	2
55	2MSQ	B	2
56	2MCQ	A	2
57	2MCQ	B	2
58	2NAT	A	2

59	2MCQ	B	2
60	2MCQ	B	2
61	2NAT	B	2
62	2MCQ	A	2
63	2MSQ	A	2
64	2MCQ	B	2
65	2MCQ	D	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Verbal | Difficulty: easy

Q2.

Answer: C

Solution:

Correct Answer: C

Subject: Engineering Mathematics | Topic: Probability | Difficulty: easy

Q3.

Answer: A

Solution:

Correct Answer: A

Subject: Genetics | Topic: Molecular Biology | Difficulty: easy

Q4.

Answer: A

Solution:

Correct Answer: A

Subject: Genetics | Topic: Mendel | Difficulty: moderate

Q5.

Answer: A

Solution:

Correct Answer: A

Subject: Biochemistry | Topic: Proteins | Difficulty: moderate

Q6.

Answer: A

Solution:

Correct Answer: A

Subject: Microbiology | Topic: Bacteria | Difficulty: difficult

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Bioprocess Engineering | Topic: Fermentation | Difficulty: moderate

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Immunology | Topic: Antibodies | Difficulty: moderate

Q9.

Answer: A

Solution:

Correct Answer: A

Subject: Bioinformatics | Topic: Sequence | Difficulty: difficult

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Cell Biology | Topic: Mitosis | Difficulty: difficult

Q11.

Answer: B

Solution:

AaBb x AaBb: phenotypic ratio 9:3:3:1.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q12.

Answer: B

Solution:

Pre-mRNA: 5 cap, intron splicing, 3 poly-A tail.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q13.

Answer: B

Solution:

|x| is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q14.

Answer: C

Solution:

Linked genes on same chromosome: recombination < 50%.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q15.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q16.

Answer: B

Solution:

Galvanic: spontaneous redox converts chemical to electrical energy.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q17.

Answer: A

Solution:

Competitive: V_{max} unchanged, K_m increases (more substrate needed).

Subject: Biology | Topic: Biology | Difficulty: difficult

Q18.

Answer: C

Solution:

$y = A \cos(2x) + B \sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q19.

Answer: B

Solution:

DNA replication: S phase in nucleus. Mitochondrial DNA replicates separately.

Subject: Biology | Topic: Biology | Difficulty: easy

Q20.

Answer: D

Solution:

Very dilute: $[H^+]$ from water dominates. $[H^+]_{total}$ approx $1.1e-7$. pH approx 6.96.

Subject: Chemistry | Topic: Chemistry | Difficulty: difficult

Q21.

Answer: A

Solution:

Reverse transcription: RNA $\rightarrow$ DNA (retroviruses like HIV).

Subject: Biology | Topic: Biology | Difficulty: easy

Q22.

Answer: A

Solution:

$$P(X \geq 2) = 1 - e^{-4}(1+4) = 1 - 5e^{-4} \approx 0.908.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q23.

Answer: D

Solution:

Perchloric acid (HClO₄) is the strongest among common acids.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q24.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q25.

Answer: B

Solution:

ABO: I^A, I^B, i (three alleles).

Subject: Genetics | Topic: Genetics | Difficulty: difficult

Q26.

Answer: A

Solution:

$$t_{1/2} = 0.693/k = 0.693/0.693 = 1 \text{ min.}$$

Subject: Chemistry | Topic: Chemistry | Difficulty: difficult

Q27.

Answer: B

Solution:

B-DNA: ~10.4 bp per turn.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q28.

Answer: B

Solution:

Light reactions: ATP + NADPH for Calvin cycle. O₂ is byproduct.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q29.

Answer: B

Solution:

$$\text{proj} = ((v \cdot w)/(w \cdot w)) \cdot w = (4/5) \cdot (2, 1, 0) = (8/5, 4/5, 0).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q30.

Answer: A

Solution:

AaBb x aabb: offspring AaBb, Aabb, aaBb, aabb in 1:1:1:1.

Subject: Genetics | Topic: Genetics | Difficulty: difficult

Q31.

Answer: B

Solution:

$$\Delta n = 2 - 4 = -2.$$

Subject: Chemistry | Topic: Chemistry | Difficulty: easy

Q32.

Answer: C

Solution:

Rolle's: f continuous on [a,b], differentiable on (a,b), f(a)=f(b). Then f'(c)=0 for some c in (a,b).

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q33.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q34.

Answer: C

Solution:

$\frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $\frac{d}{dx} \ln(\sec x) = \tan x$. $\frac{d}{dx} \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = \frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q35.

Answer: A

Solution:

$L\{t^2 f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q36.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q37.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q38.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q39.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q40.

Answer: A

Solution:

$L\{e^{(-at)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q41.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q42.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y dx - x dy = 0 \Rightarrow y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q43.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q44.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q45.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q46.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q47.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q48.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=xv$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q49.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q50.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q51.

Answer: B

Solution:

$AaBb \times AaBb$: phenotypic ratio 9:3:3:1.

Subject: Biology | Topic: Biology | Difficulty: difficult

Q52.

Answer: A

Solution:

Reverse transcription: RNA $\rightarrow$ DNA (retroviruses like HIV).

Subject: Biology | Topic: Biology | Difficulty: easy

Q53.

Answer: D

Solution:

Very dilute: $[H^+]$ from water dominates. $[H^+]_{total}$ approx $1.1e-7$. pH approx 6.96.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q54.

Answer: B

Solution:

DNA replication: S phase in nucleus. Mitochondrial DNA replicates separately.

Subject: Biology | Topic: Biology | Difficulty: easy

Q55.

Answer: B

Solution:

B-DNA: ~10.4 bp per turn.

Subject: Genetics | Topic: Genetics | Difficulty: difficult

Q56.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q57.

Answer: B

Solution:

Pre-mRNA: 5 cap, intron splicing, 3 poly-A tail.

Subject: Genetics | Topic: Genetics | Difficulty: easy

Q58.

Answer: A

Solution:

Competitive: V_{max} unchanged, K_m increases (more substrate needed).

Subject: Biology | Topic: Biology | Difficulty: easy

Q59.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q60.

Answer: B

Solution:

DNA replication: S phase in nucleus. Mitochondrial DNA replicates separately.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q61.

Answer: B

Solution:

AaBb x AaBb: phenotypic ratio 9:3:3:1.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q62.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q63.

Answer: A

Solution:

Reverse transcription: RNA → DNA (retroviruses like HIV).

Subject: Biology | Topic: Biology | Difficulty: moderate

Q64.

Answer: B

Solution:

$\Delta n = 2 - 4 = -2$.

Subject: Chemistry | Topic: Chemistry | Difficulty: difficult

Q65.

Answer: D

Solution:

Very dilute: $[H^+]$ from water dominates. $[H^+]_{total}$ approx $1.1e-7$. pH approx 6.96.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 29 questions, 51 marks
Biology: 12 questions, 19 marks
Chemistry: 8 questions, 12 marks
Genetics: 9 questions, 11 marks
General Aptitude: 1 questions, 1 marks
Biochemistry: 1 questions, 1 marks
Microbiology: 1 questions, 1 marks
Bioprocess Engineering: 1 questions, 1 marks
Immunology: 1 questions, 1 marks
Bioinformatics: 1 questions, 1 marks
Cell Biology: 1 questions, 1 marks

Difficulty Distribution:

Easy: 19 | Moderate: 32 | Difficult: 14

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.